Ring and path embedding in faulty folded Petersen cube networks

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Abstract

Many research on the folded Petersen cube networks have been published during the past several years due to its favorite properties. In this paper, we consider the ring and path embedding in faulty folded Petersen cube networks. We use $FPQ_{n,k}$ to denote the folded Petersen cube networks of parameters n and k. In this paper, we show that $FPQ_{n,k} - F$ remains hamiltonian for any $F \subseteq V(FPQ_{n,k}) \cup E(FPQ_{n,k})$ with $|F| \leq n + 3k - 2$ and $FPQ_{n,k} - F$ remains hamiltonian connected for any $F \subseteq V(FPQ_{n,k}) \cup E(FPQ_{n,k})$ with $|F| \leq n+3k-3$ if $(n, k) \notin \{(0, 1)\}\cup\{(n, 0) \mid n$ is a positive integer$\}$. Moreover, this result is optimal.

1 Introduction

A large number of form of topology have been proposed and studied for multicomputer interconnection networks. Such a form of topology is usually modeled as an undirected graph where the set of vertices represents the processors and the set of edges represents the bidirectional communication links between the processors. Existing static interconnection networks include linear arrays, rings, meshes, complete binary trees, X-trees, full-ringed binary trees, tree machines, pyramids, fat trees [15], hypercubes [17], meshes of trees, cube-connected cycles [16], de Bruijn networks [18], and so on. For general surveys on multicomputer networks, refer to [12, 13]. Among these networks, the hypercube family has been popular because of such properties as symmetry, regularity, high fault-tolerance, logarithmic degree and diameter, and selfrouting and simple broadcasting schemes. Also, several commercial multicomputer architectures (e.g., Intel iPSC/2, NCUBE/10, and connection machine CM-2) use the hypercube topology for interconnecting processors. Nevertheless, new networks are being proposed and analyzed with regard to their applicability and enhanced topological or performance properties.

It is almost impossible to design a network which is optimum from all aspects. One has to design a suitable network depending on the requirements of its properties. The hamiltonian property is one of the major requirements in designing the topology of a network. Fault-tolerance is also desirable in massive parallel systems.

In this paper, a network is represented as a loopless undirected graph. For graph definitions and notations we follow [1]. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v)$ is an unordered pair of $V\}$. We say that V is the *vertex set* and E is the *edge set.* Two vertices u and v are *adjacent* if $(u, v) \in E$. The *degree* of a vertex u of G, $\deg_G(u)$, is the number of edges incident with u. We use $\delta(G)$ to denote $\min\{\deg_G(x) \mid x \in V(G)\}$. A graph G is *k-regular* if $\deg_G(x) = k$ for any vertex in G. A *path*, $\langle v_0, v_1, v_2, \ldots, v_k\rangle$, is an ordered list of distinct vertices such that v_i and v_{i+1} are adjacent for $0 \leq i \leq k - 1$. The *length* of a path P is the number of edges in P. The *distance* between two vertices u and v in G is the length of a shortest path joining them. The *diameter* of a graph G is the distance between the farthest points in G.

A path is a *hamiltonian path* if its vertices are

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distinct and span V. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last vertex. A cycle is a *hamiltonian cycle* if it traverses every vertex of G exactly once. A graph is *hamiltonian* if it has a hamiltonian cycle. We will use K_n to denote the complete graph with n vertices and use C_n to denote the cycle graph with n vertices.

In [10], the performance of the hamiltonian property in faulty networks is discussed. A hamiltonian graph G is k *fault-tolerant hamiltonian* if $G - F$ remains hamiltonian for every $F \subset V(G) \cup E(G)$ with $|F| \leq k$. The *fault-tolerant hamiltonicity* $\mathcal{H}_f(G)$ is defined to be the maximum integer k such that G is k fault-tolerant hamiltonian if G is hamiltonian and is undefined otherwise. Clearly, $\mathcal{H}_f(G) \leq \delta(G) - 2$ if $\mathcal{H}_f(G)$ is defined. Huang et al. [10] also introduced the term, fault-tolerant hamiltonian connected. A graph G is *hamiltonian connected* if there exists a hamiltonian path joining any two vertices of G. All hamiltonian connected graphs except the complete graphs K_1 and K_2 are hamiltonian. A graph G is k *fault-tolerant hamiltonian connected* if $G - F$ remains hamiltonian connected for every $F \subset V(G) \cup E(G)$ with $|F| \leq k$. The *fault-tolerant hamiltonian connectivity* $\mathcal{H}^\kappa_f(G)$ is defined to be the maximum integer k such that G is k fault-tolerant hamiltonian connected if G is hamiltonian connected and is undefined otherwise. It can be checked that $\mathcal{H}^\kappa_f(G) \leq \delta(G) - 3$ only if $\mathcal{H}^\kappa_f(G)$ is defined and $|V(G)| \geq 4$. There are a lot of studies on fault-tolerant hamiltonicity and fault-tolerant hamiltonian connectivity [5, 6, 7, 8, 9, 10, 11, 20].

In this paper, we consider the fault-tolerant hamiltonicity and the fault-tolerant hamiltonian connectivity of the folded Petersen cube networks. The folded Petersen cube networks, proposed by [14], are motivated by the fact that the well-known Petersen graph [1], which is a 3-regular graph of 10 vertices and of diameter 2 as compared to the three-dimensional hypercube, which is a 3-regular graph of 8 vertices of diameter 3.

The k-dimensional folded Petersen network, FP_k, is constructed by an iterative Cartesian product on the Petersen graph. The folded Petersen network is then generalized into folded Petersen cube network, $FPQ_{n,k}$. The graph $FPQ_{n,k}$ defined as a product of FP_k and the n-dimensional binary hypercube, Q_n. The number of vertices in FP_k is 10^k, whereas there are $2^n \times 10^k$ vertices in $FPQ_{n,k}$, which is thus more scalable than FP_k. It turns out that $FPQ_{n,k}$ and even its special derivations $FPQ_{0,k} = FP_k$ and $FPQ_{n-3,1} = HP_n$, called the n-dimensional hyper Petersen network originally proposed by Das et al. [4], are better than the comparable-size hypercubes and several other networks with respect to the usual metrics (such as degree, diameter, connectivity, packing density, or cost) of a multicomputer architecture.

In this paper, we prove that $\mathcal{H}_f(FPQ_{n,k}) = n + 3k - 2$ and $\mathcal{H}^\kappa_f(FPQ_{n,k}) = n + 3k - 3$ if $(n, k) \notin \{(0, 1)\} \cup \{(n, 0) \mid n \text{ is a positive integer}\}$. Moreover, $FPQ_{0,1}$ is neither hamiltonian nor hamiltonian connected. Furthermore, $FPQ_{n,0}$ is hamiltonian but not hamiltonian connected if $n > 1$; $FPQ_{1,0}$ is hamiltonian connected but hamiltonian.

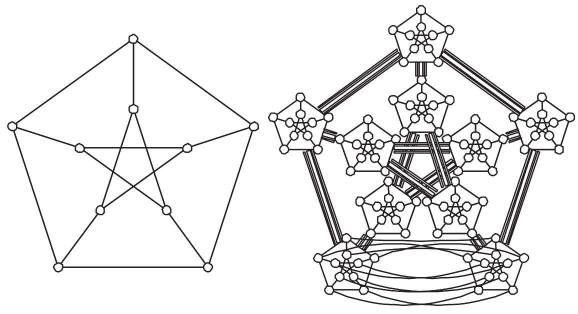
In the following section, we give the definition of the folded Petersen cube networks. In Section 3, we present some mathematical preliminary. In Section 4, we prove our main result. In the final section, we give a discussion of our work.

2 Folded Petersen cube networks

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The *cartesian product* of G_1 and G_2, denoted by $G_1 \times G_2$, is the graph with vertex set $V_1 \times V_2$ such that (u_1, v_1) is joined to (u_2, v_2) if and only if either $u_1 = u_2$ and v_1 is joined to v_2 in G_2 or $v_1 = v_2$ and u_1 is joined to u_2 in G_1. For any graph G and any positive integer k, we define $G^k = G$ if $k = 1$ and $G^k = G^{k-1} \times G$ if $k > 1$.

For any positive integer, the *n-dimensional hypercube* Q_n is defined as $Q_1 = K_2$ and $Q_n = Q_{n-1} \times Q_1$. Thus, $Q_n = K_2^n$. The *Petersen graph* P, shown in Figure 1(a), is a graph with 10 vertices has an outer 5-cycle, an inner 5-cycle, and five spokes joining them. The folded Petersen cube network $FPQ_{n,k}$ is defined as $Q_n \times P^k$. In particular, $FPQ_{0,k} = P^k$ and $FPQ_{n,0} = Q_n$. The graph $FPQ_{0,2} = P^2$ is shown in Figure 1(b).

The topology of an interconnection network plays an important role in the performance of a distributed system. Various network topologies are in use and have their respective advantages and disadvantages. A good network topology should be one with a small diameter. Regularity and symmetry are some other qualities, a good network topology is expected to possess. Some of the well-known network topologies are in use the ring topology, the hypercube, the cube-connected cycles, etc. The Petersen graph is a 3-regular graph with 10 vertices and of diameter 2. Compared to this graph, the 3-dimensional hypercube is a 3-regular graph with 8 vertices and



(a) **(b)**

Figure 1: (a) The Petersen graph P and (b) A schematic representation of P^2

of diameter 3. It has more vertices as compared to the 3-dimensional hypercube and a smaller diameter. We call it the simple Petersen graph. As an extension, Öhring and Das [14] introduce the k-dimensional folded Petersen graph, FP_k, to be $FPQ_{0,k}$. It is observed that FP_k possesses qualities of a good network topology for distributed systems with large number of sites since it accommodates 10^k vertices and is a symmetric, $3k$-regular graph of diameter $2k$. Being an iterative Cartesian product on the Petersen graph, it is scalable. Moreover, Öhring and Das [14] define the folded Petersen cube networks $FPQ_{n,k}$ and show that a number of standard topologies like linear arrays, rings, meshes, hypercubes, etc. can be embedded into it. Recently, many research on the folded Petersen cube networks have been published during the past several years due to its favorite properties [4, 14, 19].

Yet, to our knowledge, there is no study on the hamiltonian property on folded Petersen cube networks. Perhaps, the difficult is on the known result that the Petersen graph is not hamiltonian. With the result of this paper, there are only two non-hamiltonian graphs in the family of folded Petersen cube networks, namely $FPQ_{1,0}$ and $FPQ_{0,1}$.

3 Preliminary

We say a k-regular graph is *super fault-tolerant hamiltonian* if $\mathcal{H}_f(G) = k-2$ and $\mathcal{H}_f^\kappa(G) = k-3$. Some interesting families of interconnection networks are proved to be super fault-tolerant [8, 9, 10]. Chen et al. [2] observed the insight of the proofs of the aforementioned results and proposed the following construction scheme.

Let G_1 and G_2 be two graphs with the same number of vertices. Let M be an arbitrary *perfect matching* between the vertices of G_1 and G_2; i.e., M is a set of edges connecting the vertices of G_1 and G_2 in a one to one fashion. For convenience, G_1 and G_2 are called *components*. Then $G(G_1, G_2; M)$ is the graph with vertex set $V(G(G_1, G_2; M)) = V(G_1) \cup V(G_2)$ and edge set $E(G(G_1, G_2; M)) = E(G_1) \cup E(G_2) \cup M$. See Figure 2 for an illustration.

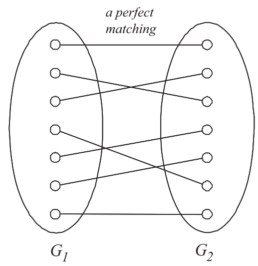
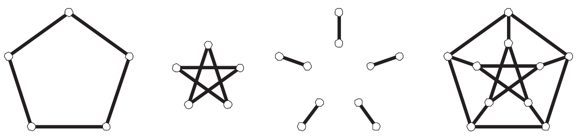


Figure 2: $G(G_1, G_2; M)$

Let H be any graph. Obviously, the cartesian product $H \times K_2$ can be viewed as $G(H, H; M)$ for some matching M. The Petersen graph P can be viewed as $G(C_5, C_5; M)$ for some matching M as shown in Figure 3.



(a) **(b)** **(c)** **(d)**

Figure 3: (a) A copy of C_5, (b) Another copy of C_5, (c) The matching M, and (d) The Petersen graph P

Furthermore, $P \times C_5$ can be viewed as $G(C_5 \times C_5, C_5 \times C_5; M)$ for some matching M. See Figure 4 for an illustration.

The following two theorems are proved in [2].

Theorem 1 *[2] Assume $k \geq 4$. Let G_1 and G_2 be two k-regular super fault-tolerant hamiltonian graphs and $|V(G_1)| = |V(G_2)|$. Then graph $G(G_1, G_2; M)$ is $(k-1)$ fault-tolerant hamiltonian.*

Theorem 2 *[2] Assume $k \geq 5$. Let G_1 and G_2 be two k-regular super fault-tolerant hamiltonian graphs and $|V(G_1)| = |V(G_2)|$. Then graph $G(G_1, G_2; M)$ is $(k-2)$ fault-tolerant hamiltonian connected.*

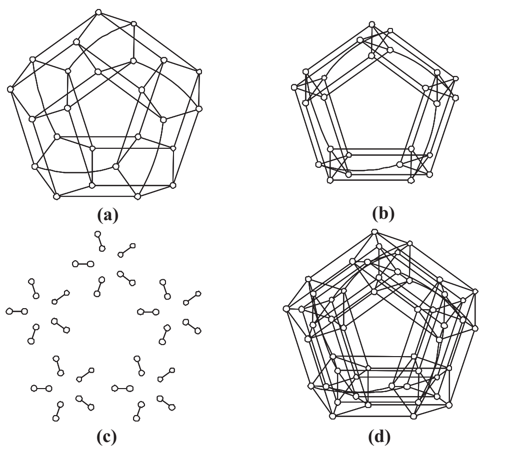


Figure 4: (a) A copy of $C_5 \times C_5$, (b) Another copy of $C_5 \times C_5$, (c) The matching M, and (d) $P \times C_5$

Combining Theorems 1 and 2, we have the following Corollary.

Corollary 1 *Assume $k \geq 5$. Let G_1 and G_2 be two k-regular super fault-tolerant hamiltonian graphs and $|V(G_1)| = |V(G_2)|$. Then graph $G(G_1, G_2; M)$ is $(k+1)$-regular super fault-tolerant hamiltonian connected.*

Chen et al. [3] further extend their work by considering another construction scheme. Let r and t be positive integers with $r \geq 3$. Assume that $G_0, G_1, \cdots, G_{r-1}$ are graphs with $|V(G_i)| = t$ for $0 \leq i \leq r-1$. We define $H = G(G_0, G_1, \cdots, G_{r-1}; \mathcal{M})$ with $V(H) = \bigcup_{i=0}^{r-1} V(G_i)$ and $E(H) = \mathcal{M} \cup \bigcup_{i=0}^{r-1} E(G_i)$, where $\mathcal{M} = \bigcup_{i=0}^{r-1} M_{i,i+1 \; (\mathbf{mod}\; r)}$ with $M_{i,i+1 \; (\mathbf{mod}\; r)}$ is any arbitrary perfect matching between $V(G_i)$ and $V(G_{i+1 \; (\mathbf{mod}\; r)})$. See Figure 5 for an illustration. Let H be any graph. Obviously, the cartesian product $H \times C_4$ can be viewed as $G(H, H, H, H; \mathcal{M})$. The following theorem is proved in [3].

Theorem 3 *[3] Assume that $G_0, G_1, \ldots, G_{n-1}$ are k-regular super fault-tolerant hamiltonian with the same number of vertices where $n \geq 3$ and $k \geq 5$. Then $G(G_0, G_1, \ldots, G_{n-1}; \mathcal{M})$ is $(k+2)$-regular super fault-tolerant hamiltonian.*

Theorem 1, Theorem 2, and Theorem 3 are useful to construct super fault-tolerant hamiltonian graphs. Yet, the drawback of these theorems are the restriction of k are rather high. However, we have some difficulty to improve Theorem 2 by including the case $k = 4$. For this reason, in this paper we introduce the concept of extendable 4-regular super fault-tolerant hamiltonian graph. A 4-regular super fault-tolerant hamiltonian graph H is *extendable* if $H - \{x, y\}$ remains hamiltonian connected for any x and y such that $(x, y) \in E(H)$.

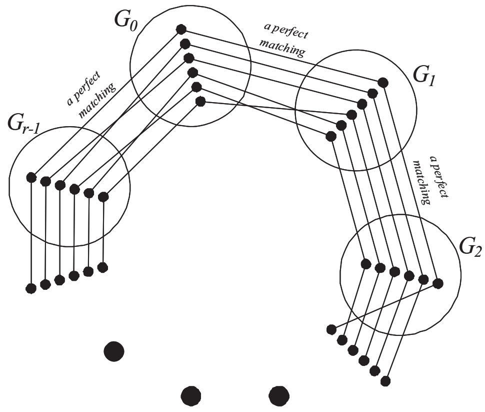


Figure 5: $H = G(G_0, G_1, \ldots, G_{r-1}; M)$

The following lemma is proved by brute force. Here we just state the result for reducing the complexity.

Lemma 1 *Both $P \times Q_1$ and $C_5 \times C_5$ are extendable 4-regular super fault-tolerant hamiltonian graphs. However, $C_5 \times C_4$ is a 4-regular super fault-tolerant hamiltonian but not extendable graph.*

Theorem 4 *Assume that G_1 and G_2 are extendable 4-regular super fault-tolerant hamiltonian graphs with $|V(G_1)| = |V(G_2)|$. Then $G(G_1, G_2; M)$ is 5-regular super fault-tolerant hamiltonian.*

4 Main result

Lemma 2 *P^k is $(3k)$-regular super fault-tolerant hamiltonian if and only if $k \geq 2$.*

Theorem 5 *$\mathcal{H}_f(FPQ_{n,k}) = n + 3k - 2$ and $\mathcal{H}_f^\kappa(FPQ_{n,k}) = n + 3k - 3$ if $(n, k) \notin \{(0, 1)\} \cup \{(n, 0) \mid n$ is a positive integer$\}$. Moreover, $FPQ_{0,1}$ is neither hamiltonian nor hamiltonian connected. Furthermore, $FPQ_{n,0}$ is hamiltonian but not hamiltonian connected if $n > 1$; $FPQ_{1,0}$ is hamiltonian connected but not hamiltonian.*

5 Discussion

We believe that our approach of this paper can be applied to the same problem on other interconnection networks. Definitely, we can repeatedly apply Theorem 1, Theorem 2, and Theorem 3 to obtain the result in this paper. However, we need a lot of efforts to check the base cases for the requirement $k \geq 5$ in Theorem 2 and Theorem 3. By introducing the concept of extendable 4-regular super fault-tolerant hamiltonian graph, all difficulties are solved immediately. Thus, it would be a great improvement if Theorem 1, Theorem 2, and Theorem 3 remain true for smaller k.

Let H be the graph shown in Figure 6(a). Obviously, H is 3-regular graph. By brute force, we can check that H is 3-regular super fault-tolerant hamiltonian. Let G_1 and G_2 be two copies of H. Let G be the graph shown in Figure 6(b). Obviously, $G = (G_1, G_2; M)$ for some matching M. By brute force, we can prove that $G - \{2, 12\}$ is not hamiltonian. Thus, G is not 2 fault-tolerant hamiltonian.

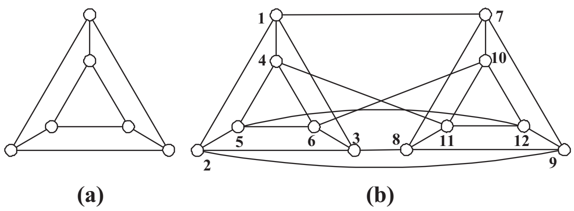


Figure 6: (a) The graph H and (b) The graph G

Hence, the bound of k in Theorem 1 is optimal. Yet, we believe the bounds of k in Theorem 2 and Theorem 3 to be also optimal.

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